MSG No. 50.279 $Pb'an$ [Type III, orthorhombic]

Table 1: Wyckoff site: `2a`, site symmetry: `22'2'`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[1,2,5,6]`
2	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[3,4,7,8]`

Table 2: Wyckoff site: `2b`, site symmetry: `22'2'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[1,2,5,6]`
2	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[3,4,7,8]`

Table 3: Wyckoff site: `2c`, site symmetry: `22'2'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[1,2,5,6]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[3,4,7,8]`

Table 4: Wyckoff site: `2d`, site symmetry: `22'2'`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[1,2,5,6]`
2	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[3,4,7,8]`

Table 5: Wyckoff site: `4e`, site symmetry: `-1'`

No.	position	mapping
1	$[0, 0, 0]$	`[1,7]`
2	$\left[0, \frac{1}{2}, 0\right]$	`[2,8]`
3	$\left[\frac{1}{2}, 0, 0\right]$	`[3,5]`
4	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[4,6]`

Table 6: Wyckoff site: `4f`, site symmetry: `-1'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,7]`
2	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[2,8]`
3	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[3,5]`
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[4,6]`

Table 7: Wyckoff site: `4g`, site symmetry: `2..`

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ 0\right]$	`[1,2]`
2	$\left[x+\frac{1}{2},\ \frac{3}{4},\ 0\right]$	`[3,4]`
3	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ 0\right]$	`[5,6]`
4	$\left[-x,\ \frac{3}{4},\ 0\right]$	`[7,8]`

Table 8: Wyckoff site: `4h`, site symmetry: `2..`

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ \frac{1}{2}\right]$	`[1,2]`
2	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{2}\right]$	`[3,4]`
3	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{1}{2}\right]$	`[5,6]`
4	$\left[-x,\ \frac{3}{4},\ \frac{1}{2}\right]$	`[7,8]`

Table 9: Wyckoff site: `4i`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[\frac{1}{4},\ y,\ 0\right]$	`[1,5]`
2	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ 0\right]$	`[2,6]`
3	$\left[\frac{3}{4},\ -y,\ 0\right]$	`[3,7]`
4	$\left[\frac{3}{4},\ y+\frac{1}{2},\ 0\right]$	`[4,8]`

Table 10: Wyckoff site: `4j`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[\frac{1}{4},\ y,\ \frac{1}{2}\right]$	`[1,5]`
2	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[2,6]`
3	$\left[\frac{3}{4},\ -y,\ \frac{1}{2}\right]$	`[3,7]`
4	$\left[\frac{3}{4},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[4,8]`

Table 11: Wyckoff site: `4k`, site symmetry: `..2'`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ z\right]$	`[1,6]`
2	$\left[\frac{1}{4},\ \frac{1}{4},\ -z\right]$	`[2,5]`
3	$\left[\frac{3}{4},\ \frac{3}{4},\ z\right]$	`[3,8]`
4	$\left[\frac{3}{4},\ \frac{3}{4},\ -z\right]$	`[4,7]`

Table 12: Wyckoff site: `4l`, site symmetry: `..2'`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$	`[1,6]`

continued ...

Table 12

No.	position	mapping
2	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	[2,5]
3	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	[3,8]
4	$\left[\frac{3}{4}, \frac{1}{4}, -z\right]$	[4,7]

Table 13: Wyckoff site: 8m, site symmetry: 1

No.	position	mapping
1	$[x, y, z]$	[1]
2	$\left[x, \frac{1}{2} - y, -z\right]$	[2]
3	$\left[x + \frac{1}{2}, -y, z\right]$	[3]
4	$\left[x + \frac{1}{2}, y + \frac{1}{2}, -z\right]$	[4]
5	$\left[\frac{1}{2} - x, y, -z\right]$	[5]
6	$\left[\frac{1}{2} - x, \frac{1}{2} - y, z\right]$	[6]
7	$[-x, -y, -z]$	[7]
8	$\left[-x, y + \frac{1}{2}, z\right]$	[8]